

DIFFA, Niger

WMO: 610850

Lat: 13.373N

Lon: 12.627E

Elev: 994

StdP: 14.18

Time Zone: 1.00 (E01)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	54.5	57.0	18.0	14.1	76.2	20.0	15.6	76.9	25.6	67.4	22.1	69.3	2.0	40	0.398

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	26.4	109.5	71.2	107.8	71.1	105.9	71.2	80.6	92.2	79.8	91.6	79.1	91.0	5.9	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.9	150.6	85.6	77.1	146.3	84.8	76.3	142.6	84.3	45.0	91.8	44.1	91.7	43.4	91.4	83.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.1	17.8	15.6	DB	50.3	112.2	1.9	0.6	48.9	112.6	47.8	112.9	46.8	113.2	45.4	113.6	(4)
(5)			WB	40.6	82.4	0.9	0.5	39.9	82.8	39.3	83.1	38.8	83.4	38.1	83.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	84.0	72.3	77.7	85.0	91.8	93.8	91.7	86.6	82.6	85.7	87.0	80.9	73.2		
	DBStd	7.77	4.65	5.55	4.26	3.77	3.14	4.04	4.04	3.23	3.13	2.52	3.62	3.99		
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD65	1	1	0	0	0	0	0	0	0	0	0	0	0		
	CDD50	12425	690	776	1084	1253	1359	1252	1135	1012	1071	1147	926	720		
	CDD65	6952	227	356	619	803	894	802	670	547	621	682	476	255		
	CDH74	92096	2271	4072	8353	12282	13905	11999	8838	5944	7535	8888	5544	2465		
	CDH80	55036	987	2160	5171	8402	9637	7811	4820	2579	3965	5319	3147	1038		
(14) Wind	WSAvg	5.1	5.8	5.6	5.0	3.9	4.9	6.3	6.4	4.3	4.1	3.9	4.8	6.1		
(15) Precipitation	PrecAvg	11.3	0.0	0.0	0.0	0.1	0.2	1.0	3.5	4.7	1.7	0.2	0.0	0.0		
	PrecMax	18.9	0.0	0.2	0.2	0.5	1.8	3.7	8.0	8.6	3.8	1.0	0.0	0.0		
	PrecMin	7.0	0.0	0.0	0.0	0.0	0.0	0.1	1.4	1.6	0.1	0.0	0.0	0.0		
	PrecStd	3.3	0.0	0.0	0.0	0.1	0.4	1.0	1.9	1.6	0.9	0.3	0.0	0.0		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	98.1	103.7	108.1	111.3	111.6	108.9	104.1	97.8	103.7	105.1	101.9	97.5		
		MCWB	64.6	67.3	69.2	70.7	72.4	74.1	75.8	78.6	74.3	70.6	66.4	66.0		
	2%	DB	93.9	100.0	105.3	109.5	109.7	106.5	100.8	94.8	100.7	103.2	99.6	92.7		
		MCWB	61.8	64.3	67.8	70.0	72.0	74.6	76.1	79.0	75.8	70.7	66.5	62.6		
	5%	DB	89.9	96.0	102.4	107.7	108.1	104.3	98.1	92.7	98.2	101.4	97.2	89.7		
		MCWB	59.9	62.6	65.5	69.2	71.9	74.9	76.6	78.7	76.7	70.8	65.6	61.0		
	10%	DB	85.9	92.3	99.3	105.4	106.2	101.8	95.6	90.7	95.4	99.1	94.5	86.2		
		MCWB	58.4	60.9	64.5	68.9	72.0	75.3	76.7	78.5	77.5	70.7	64.6	59.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	66.2	70.4	76.2	76.8	79.9	79.8	80.7	81.7	81.7	80.1	70.6	67.0		
		MCDB	95.2	98.4	99.2	103.5	103.1	97.2	93.9	91.8	93.2	93.3	95.5	94.6		
	2%	WB	62.9	65.6	71.3	75.1	78.1	78.9	79.7	80.6	80.4	78.6	68.5	64.2		
		MCDB	90.9	96.0	99.1	100.9	100.5	96.8	92.6	90.3	91.9	91.9	94.8	89.7		
	5%	WB	60.6	63.6	68.1	73.7	77.0	78.1	78.8	79.8	79.6	77.3	66.7	62.0		
		MCDB	87.8	93.7	97.1	98.8	99.3	95.7	91.2	89.1	91.1	90.8	93.5	86.6		
	10%	WB	58.8	61.7	65.7	72.0	75.9	77.3	78.0	79.0	78.8	76.0	65.3	60.4		
		MCDB	84.6	90.4	96.5	97.9	97.6	94.5	90.1	88.0	90.4	89.5	92.1	84.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	28.6	29.3	29.7	28.9	26.4	20.7	17.4	15.4	19.9	26.3	29.3	27.9		
		MCDBR	33.9	33.2	34.0	31.2	28.9	23.6	20.3	18.3	23.8	29.5	31.6	32.0		
		MCWBR	15.3	13.9	13.8	12.0	11.1	7.3	5.3	5.8	6.8	9.5	13.2	14.6		
		MCDBR	33.1	32.1	30.4	26.7	23.9	20.3	17.7	16.9	19.5	22.0	29.7	30.0		
	5% WB	MCWBR	15.8	14.2	14.0	11.4	8.9	6.6	5.8	6.7	7.1	8.0	13.9	14.6		
(40) Clear-Sky Solar Irradiance	taub		0.555	0.618	0.665	0.745	0.725	0.707	0.674	0.624	0.628	0.660	0.579	0.574		
	taud		1.701	1.571	1.493	1.400	1.451	1.535	1.650	1.757	1.729	1.595	1.655	1.665		
	Ebn at Noon		236	227	221	204	204	205	213	227	227	215	227	226		
	Edh at Noon		74	88	97	106	99	90	80	73	75	84	77	75		
(44) All-Sky Solar Radiation	RadAvg		1812	1951	2084	2133	2116	2010	1869	1789	1943	1929	1858	1751		
	RadStd		46	46	91	72	88	86	53	83	41	59	50	46		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page